

ANNEXURE-3 List of minimum Tools & Equipments for a batch of 30 students			
Qualification Title : REFRIGERATION & AIR CONDITIONING			Sector : Construction
NSQF Level : Level-4			Notional Hours: 2880
SL NO	NAME OF THE TOOL & EQUIPMENT	SPECIFICATION	QUANTITY
STANDARD TOOL KIT (GENERAL WORKSHOP)			
1	Anvil	150 mm blade	1 Nos.
2	Auto darkening welding helmet	150 mm	15 Nos.
3	Bearing different types		6 Nos.
4	Bench vice	8 inch	12 Nos.
5	Bend Snips	250 mm	15 Nos.
6	Blow Lamp	300 mm	6 nos.
7	C- Clamp	300 mm	12 Nos.
8	Caliper inside spring type	300 mm	12 Nos.
9	Caliper Outside spring type	250 mm	6 nos.
10	Centre Punch	150 mm	3 nos.
11	Chipping hammer	Set of 12, Length = 150 mm	1 set
12	Chisel cold	220 g with handle (1/2 lb)	6 Nos.
13	Clamp "C"	6 mm	2 set
14	Combination Set	6mm to 34mm	2 set
15	Conductor stake	300 mm	2 nos.
16	Copper smith stake	0 - 25 mm	2 nos.
17	Different Profile gauges (Plate type) - For		1 no.
18	Digital Micrometer outside.	12 mm driver, sized10-32 mm set of 18	1 set
19	Digital Vernier caliper.	100 mm	2 nos.
20	Divider	200 mm	2 nos.
21	Divider spring type	Nominal Dia. 6, 8, 10, 12,16mm	1 set
22	Dot Punch	12 - 16mm set of 5.	1 no.
23	Double End spanner set	Standared size	6 nos.
24	Double Ended tubular Box spanner set with Tommy	Standared size	2 nos.
25	Dowel pin (straight)	150 mm	6 nos.
26	Earth clamp	100 mm	1 no.
27	Electrode holder	Dia. -1" Length -4" (Mat: Stainless Steel)	1 each
28	Engineer's try square (knife-edge)	M3, M4, M5, M6, M8, M10, M12, M14,	1 each
29	Feeler gauge	Dia. -6"	2 each
30	File card / Wire Brush	M6, M8, M10, M12	2 each
31	File flat - bastard	150NB, Class# 150 Flange:	2 nos.
32	File flat - Second cut	M20x2.5x90Long (part thread - ex.Head)	15 nos.
33	File flat - Smooth	Ratchet type Die head of 1/2", 3/4" and 1"	2 nos.
34	File flat second cut safe edge		1 no.
35	File half round - Second cut	to fit on 50mm dia. Shaft with key slot	1 no.
36	File half round bastard.	to fit with key slot of shaft & pulley	2 nos.
37	File hand second cut	to fit 50mm dia. Shaft	2 nos.
38	File knife edge smooth	to fit on pulley	1 each
39	File round - Second cut	PTFE gasket roll small size	1 no.
40	File round bastard	minimum 25mm size, carbon steel material	2 each
41	File square second cut	Straight Shank Hollow Punch Sets 5-12mm	1 set
42	File Swiss type needle	200mm hardened and black finish	2 nos.
43	File triangular - Smooth	Each type of diameter 25mm (min.)	1 each
44	File triangular - second cut	8mm Nominal Dia. Single leg sling	2 nos.
45	First aid box	Universal gear puller 2 or 3 jaws adjustable	1 no.
46	Flat file	25 to 50 mm by 5 mm, Set of 6 pieces	1 Set

SL NO	NAME OF THE TOOL & EQUIPMENT	SPECIFICATION	QUANTITY
47	Funnel stake	2 Pounds	1 No.
48	Gas lighter	8 - 150 mm	1 no.
49	Gauge plug		10 no.
50	Hacksaw frame fixed type	125 mm	1 no.
51	Hammer Ball peen	0 - 25 mm L.C. 0.001 mm.	10 no.
52	Hammer Ball peen	500 gm	3 no.
53	Hollow punch	250 gm	3 nos.
54	Inside Caliper	150 X 150 X 250 mm	2 nos.
55	Interchangeable ratchet socket set	Whitworth	1 set
56	Letter punch	Standard	1 set
57	Mallet	8"	10 Nos.
58	Marking table (Mild steel)	8"	6 Nos.
59	Micrometer	2"	4 Nos.
60	Micrometer outside.	6"	2 nos.
61	Number punch	standard	1 set
62	Oil Can	250 ml	2 nos
63	Pliers combination insulated	250 mm	4 nos.
64	Pocker	200cm x 150cm x 45 cm	1
65	Portable hand drill (Electric)		1
66	Prick punch	with 8 Pigeon holes	2
67	Safety goggles	150 mm	6 Nos.
68	Scraper flat	200 mm	6 Nos.
69	Screw driver	Fixed (300 mm)	15 Nos.
70	Screw driver	flat 19mm x 150mm	6 Nos.
71	Scriber	250 gm with metal handle	6 Nos.
72	Sledge hammer	500 gm	2 Nos.
73	Spanner adjustable		2 nos
74	Spanner- Double Ended	Standard size	6 nos.
75	Spindle key for cylinder valve.	Double faced, 4 kg	2 nos.
76	Stainless Steel Rule	6" heavy duty	15 Nos.
77	Standard Tap screws		2 Nos.
78	Standard Wire Gauge		1 No
79	Steel lockers		2 Nos.
80	Straight Snips		10 Nos.
81	Surface Gauge		2 nos.
82	Surface Plate	12 sq inches working area with stand	1 no.
83	Trolley for gas cylinders.	6mm to 34mm	2 Nos.
84	Try Square		4 Nos.
85	Universal Surface gauge		2 Nos.
86	Universal weld measuring gauge		2 Nos.
87	V-Block		1 each
88	Vernier caliper	6 mm x 75 mm HSS/Carbide	1 nos.
89	Vernier Height Gauge	10 mm x 75 mm HSS/Carbide	1 nos.
90	Welding goggles	6" heavy duty	2 nos.
91	Welding hand shield fiber	150x100x100 mm	1 no.
92	Welding helmet fiber	250 mm arm	2 no.
93	Welding helmet with coloured flame	750 gm with handle	2nos.
94	Welding torches.	inch and mm 8"/200 mm. LCM	1 no.
95	Wire brush	(0 to 25 mm & 25 to 50 mm) LC 0.001 mm.	1 no.
96	Wire gauge.	Standard	1 set
97	Wooden handle forged Soldering Iron copper bit.		1 set
REFREGERATION AND AIR CONDITION TOOLS			
19	Pipe flaring tool	3/16" to 1/2"	10 Nos.

SL NO	NAME OF THE TOOL & EQUIPMENT	SPECIFICATION	QUANTITY
20	Pipe flaring tool	5/8" to 3/4"	10 Nos.
21	Pipe flaring Tool double flare	3/16 to 1/2" cap	10 Nos.
22	Tube cutters	1/8" to 1" Dia	10 Nos.
23	Measuring tape steel	6 ft	05 Nos.
24	Soldering Iron Electric	25 watts with round pencil bit	10 Nos.
25	Soldering Iron	Electric 65 watts with oval bit	04 Nos.
26	Compound gauge	21/2"dia. with 11b gradation 30 to 150 lbs	04 Nos.
27	Pressure gauge	21/2 dia 0 to 300 lbs with 5 lb graduation	04 Nos.
28	Shut off value socket and tee handled key	1/4"	04 Nos.
29	Gland packing socket and tee handled key	1/4"	04 Nos.
30	Gland packing socket and tee handled key	3/16"	04 Nos.
31	Refrigeration cylinders	F 12 with key 5 lbs cap	1 set
32	A pocket testing lamp	Standared	10 Nos.
33	Charging line	1/4" copper tube 25 inch	04 Nos.
34	Seal seat puller Universal		2 Nos.
35	Hallow punches	1/4" to 1" set of eight pieces	02Nos.
36	Oil can pressure type	1/8 pint cap	04 Nos.
37	Swaging tools	1/4" to 3/8" set	6 Nos.
38	Spring balance dial type	0-20 kg	1 Nos
39	Feeler gauges	8" long 1/2" wide tapered to 1/4" 26 blades	6 Nos.
40	Pipe bender	spring types for Round pipe Universal set	4 Nos.
41	Ratchet Wrench	1/16" and 1/4 " set	4 Nos.
42	Brazing torch	Normal sixe	10 Nos.
43	Flaring block adjustable	Standared size	10 Nos.

SL NO	NAME OF THE TOOL & EQUIPMENT	SPECIFICATION	QUANTITY
REFREGERATION AND AIR CONDITION EQUIPMENTS			
44	Portable Forge with blower	with all accessories (400A, OCV 60-100V. 60% duty cycle)	1 set
45	Refrigerator House hold assorted types (Reciprocating) and with different capacities	with all accessories (300A, OCV 60-100V. 60% duty cycle)	1 set
46	Compressor, Fractional H.P. for Refrigerator	400A, OCV 60-100V. 60% duty cycle	2 Nos.
47	Air conditioner window type 1 Ton	1 Ton with R22 reciprocating	2 Nos.
	Air conditioner split	1 Ton with R22/ 134 a reciprocating	2 Nos
48	Refrigerator, Absorption type	with water cooled torch 300A, Argon regulator, Gas hose, water circulating system and standard accessories	1 Nos.
49	Air compressor with 65-70 gallons cap	capable of cutting straight and circular with all accessories	2 Nos.
50	Thermostatic expansion valve	150 mm dia	5 Nos.
51	Automatic expansion valve	For Refrigeration and AC	5 Nos.
52	Thermostatic switch	14" dia wheel	5 Nos.
53	Low pressure switch(domestic)	with fire bricks	5 Nos.
54	Solenoid Valve	with positioner	5 Nos.
55	High pressure (switch domestic)	Refrigerator	5 Nos.
56	Relay for Air conditioner	Airconditioning	5 Nos.
57	Operating switch for Air conditioner	Airconditioning	5 Nos.
58	Cheek value	cut 6mm sheets and flats	5 Nos.
59	Temperature valve	14"	5 Nos.
60	Liquid line indicator 3/8" X 3/8"	Capacity 6mm	5 Nos.
61	Refrigerant Filters	0 to 350 degree C, 10kg capacity	5 Nos.
62	Two way valve 1/4" X 1/4"	Standared	5 Nos.
63	Three way valves 1/4" X 1/4"	Standared	5 Nos.
64	Car Air-conditioning unit		1 Nos
65	Air conditioner inverter Split type	1 Ton capacity	2 nos.
66	Two stage rotary vacuum pump		1Nos
67	Gas leak detector for halogen gas	12 sq inches working area with stand	1Nos
68	Electronic leak detector		1Nos
69	Fin comb		4 Nos
70	Frost free refrigerator	200 Litre miniimum	1Nos
71	Refrigerant cylinder	Foam type and CO2 type	2 Nos (Min)
72	Car AC cut section model		1 Nos.
73	D.C. Volt Meter	0-500 v	4 Nos
74	D.C. Ammeter	0-25 A	4 Nos
75	D.C. Volt meter	0-500 V	4 Nos
76	A.C. Ammeter	0-25 A	4 Nos
77	Multi meter	Digital	4 Nos
78	Clamp meter	Digital	4 Nos
79	Watt meter	1 kw	4 Nos
80	Water cooler		1 Nos
81	Working model of Ice Candy plant.		1 Nos
82	Chiller unit.		1 Nos
83	Cooling tower model		1 Nos
84	VRV and VRF models.		1 Nos
85	Centralised AC unit set up.		1 Nos

Common facility for all programmes in an Institution			
CLASS ROOM FURNITURE			
1	Instructors table		1 no.
2	Instructors chair		1 no.
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Almirah		1 no
5	Black/White board	1200 x 900 mm	1 no.
6	Instructors Laptop with internet facility	Windows with MS office package	1 no.
7	LCD Projector		1 no
IT LAB *			
1	Table & Chair for Instructors		Sufficient
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Steel almarah / cup board		1 no.
5	Black/White board	1200 x 900 mm	1 no.
6	Desktop with internet facility	Windows with MS office package (Latest configuration)	10 nos.
7	LCD Projector		1 no
8	Printer	A4 Size	1 no.
9	Computer table		Sufficient number for arranging 10 computers

* Common for up to 5 programmes or 5 batches in an Institution. For Institutions with more than 5 programmes or 5 batches there shall be proportionate increase in number of computers and related furniture.